

A Multimodal IoT Scene Narrator Integrating YOLOv8 and GPT-4V for Real-Time Audio Scene Understanding

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Abstract

We present the Multimodal IoT Scene Narrator, a real-time hybrid pipeline that integrates YOLOv8 object detection with the GPT-4V vision-language model to produce natural-language audio scene descriptions for visually impaired users. The system serves as a software proof-of-concept for the Environmental Awareness module of the Beyond Vision assistive eyeglasses project. Our architecture decouples high-frequency spatial perception from periodic contextual reasoning: YOLOv8 runs on every camera frame to ground the language model with structured detection metadata, while GPT-4V is queried every five seconds with both the raw frame and the detection summary to generate concise GPS-style narration, vocalized via OpenAI TTS with offline pyttsx3 fallback. We evaluated the system across two recorded sessions on a commodity laptop containing 54 narration events spanning single-room, multi-room walkthrough, person, animal, and occlusion conditions. The pipeline produced human-like, scene-grounded narrations in 53 of 54 events, with a median end-to-end VLM latency of 2.81 seconds. Our analysis surfaces three findings that complicate the standard hybrid narrative. First, YOLO produced zero detections on 42.6 percent of frames in normally-lit indoor scenes, meaning the VLM carried the entire perceptual load on nearly half of all frames. Second, in every observed instance of YOLO categorical error — a phone-screen cat-wallpaper, a false-positive toilet, a phantom person, and others — GPT-4V rejected or hedged on the detector’s label rather than echoing it, making the corrective behavior bidirectional. Third and most consequentially for assistive use, the system gave actionable navigation guidance in only 2 of 54 events; the remaining narrations were descriptive rather than directive, which we argue is the central practical gap between current VLM-based scene narration and deployable assistive technology. We discuss the implications for the Beyond Vision hardware port and propose prompt-level and architectural mitigations.

Keywords—Vision-Language Models, YOLOv8, GPT-4V, Object Detection, Internet of Things, Multimodal AI, Assistive Technology, Real-Time Scene Narration, Text-to-Speech, Edge Deployment

I. INTRODUCTION

Approximately 43 million people worldwide live with complete blindness and over 295 million experience moderate to severe visual impairment [1]. For these individuals, understanding the immediate environment remains a daily challenge that conventional assistive devices such as white canes and screen readers address only in fragmented ways [2]. The rapid maturation of Vision-Language Models (VLMs) over the past three years has opened a fundamentally new design space for assistive scene narration, in which a single neural system can perceive a visual scene and articulate it in natural language [3], [4].

However, deploying VLMs directly as the sole perception system has well-documented limitations. Monolithic VLMs such as GPT-4V exhibit two- to five-second cloud inference latencies, suffer from object hallucination, and provide only weak spatial grounding [5], [6]. Conversely, traditional object detectors in the YOLO family achieve sub-100 ms inference and precise localization but produce only categorical labels with no contextual reasoning [7], [8]. Our prior survey of more than 30 systems published between 2021 and 2025 identified a clear gap: few practical implementations combine the spatial precision of YOLO with the contextual richness of VLMs in a unified, real-time pipeline suitable for assistive applications [9].

This paper presents the Multimodal IoT Scene Narrator, an implemented hybrid pipeline that addresses this gap. The system runs YOLOv8-nano continuously on every webcam frame for low-latency object detection, and every five seconds invokes GPT-4V with both the raw frame and a structured summary of the YOLO detections to produce a concise, GPS-style scene narration that is spoken aloud via text-to-speech (TTS). The work serves a dual purpose: as a standalone demonstration of practical multimodal VLM-IoT integration on commodity hardware, and as a software-level proof-of-concept for the Environmental Awareness module of the Beyond Vision assistive eyeglasses platform.

The contributions of this work are fourfold:

- 1) A complete, open implementation of a hybrid YOLOv8 + GPT-4V + TTS pipeline that runs on a standard laptop and produces real-time audio scene narration.
- 2) An empirical evaluation across two recorded sessions totaling 54 narration events, covering single-room, multi-room walkthrough, person, animal, and full and partial occlusion conditions, with per-event latency instrumentation in the second session.
- 3) Empirical evidence that the value of hybridization is bidirectional: YOLO grounds the VLM on objects present, but the VLM also disambiguates and corrects YOLO’s categorical errors and supplies the bulk of user-facing utility on the 42.6 percent of frames where the detector is silent.
- 4) Identification of a navigation-guidance gap: across 54 events the system produced actionable directive guidance only twice, with the remaining narrations being descriptive rather than directive. We argue this is the central practical gap for deployable assistive use and propose targeted mitigations.

The remainder of this paper is organized as follows. Section II reviews related work in VLM-based assistive narration and hybrid detection architectures. Section III details the system architecture and design. Section IV describes the implementation. Section V presents the experimental setup. Section VI reports results. Section VII discusses limitations and future work, and Section VIII concludes.

II. RELATED WORK

A. VLMs for Assistive Scene Description

The integration of GPT-4V into the Be My Eyes platform in 2023 brought VLM-based assistive narration to mainstream attention, demonstrating that conversational, image-grounded assistance could help blind and low-vision (BLV) users with diverse daily tasks [3], [10]. Envision Glasses extended this paradigm into wearable form, embedding scene description into smart eyewear [11]. NaviGPT focused specifically on brief ahead-looking navigation narration on mobile

devices [12], while LifeInsight explored question-answering over an egocentric video stream using GPT-4 [13]. A diary study by Gonzalez Penuela et al. with 16 BLV participants found that users regularly relied on AI scene description apps despite occasional errors, indicating strong real-world utility [14].

B. Hybrid Detector-VLM Architectures

Alam et al. provided the first comprehensive comparison of large VLMs with YOLO, SSD, and Faster R-CNN for object detection, and advocated for hybrid frameworks that fuse contextual reasoning with detector-level spatial precision [6]. Kim et al. instantiated this idea by combining YOLOv8s with the Florence-2 VLM for fire detection in enclosed parking environments, demonstrating that VLM-based contextual verification could significantly reduce false positives produced by the detector alone [15]. In remote sensing, recent work has shown that YOLO bounding boxes can be used to narrow the VLM’s attention to relevant image regions, suppressing background clutter and reducing hallucination [16]. The Generalist YOLO framework further blurred the line between detector and VLM by building a unified encoder for both tasks, achieving 42× faster inference than comparable generalist multimodal models [17].

C. Wearable Assistive IoT

Wearable assistive devices have progressed from simple ultrasonic attachments to AI-powered systems with tightly integrated sensors and processors. Lokesh et al. developed an ESP32-CAM and Arduino-based device using YOLOv4 for detection and SVM for face recognition, reporting greater than 90 percent accuracy with low latency [18]. Gao et al. built smart glasses with cross-modal learning for obstacle avoidance, achieving response times below 320 ms and a battery life exceeding ten hours, though their prototype remained bulky at approximately 400 g [19]. Commercial products such as OrCam MyEye have demonstrated user demand for clip-on assistive vision [20]. The recent Binghamton University GPT-4 robot guide dog illustrated how VLM-driven scene verbalization can extend to robotics platforms as well [21].

D. Positioning of This Work

Our system differs from prior cloud-only assistive narrators (Be My Eyes, Envision, NaviGPT) by integrating a continuous on-device detector that grounds VLM prompts, and it differs from detector-only wearables by incorporating contextual VLM narration. It most closely resembles the YOLO+Florence-2 fire detection pipeline of Kim et al. [15], but targets the broader and more open-ended task of general indoor and outdoor scene description for assistive use rather than safety-critical anomaly detection.

III. SYSTEM ARCHITECTURE AND DESIGN

A. Architectural Overview

The Scene Narrator follows a three-stage hybrid pipeline. Stage one is continuous, low-latency object detection: every captured webcam frame is passed through YOLOv8-nano, which produces bounding boxes, class labels, and confidence scores. Stage two is periodic vision-language narration: every five seconds, the most recent frame and a textual summary of the current YOLO detections are sent together to GPT-4V, which returns a short natural-language description tailored to a visually impaired listener. Stage three is audio output: the returned narration is passed to a TTS engine (OpenAI TTS by default, with offline pyttsx3 fallback) and played through the system audio. The data flow is illustrated conceptually in Fig. 1.

Fig. 1. Three-stage hybrid pipeline: continuous YOLO detection feeds periodic GPT-4V narration, which drives TTS audio output.

B. Design Rationale

Three design decisions shaped the architecture. First, decoupling detection frequency from narration frequency was essential: detection must run on every frame so that the bounding-box overlay and potentially future obstacle alerts remain responsive, but invoking a cloud VLM at 30 frames per second would be both economically infeasible and acoustically incoherent. A five-second narration interval was chosen empirically as a balance between providing fresh scene context and not overwhelming the listener with continuous speech.

Second, the YOLO detection summary is passed to GPT-4V as structured text in the prompt, not just as a raw image. This is the core hybrid mechanism: the language model receives both the pixels and a verified list of objects, which anchors its generation and reduces the chance of confabulating absent items. This decision is directly informed by the findings of Kim et al. [15] and Alam et al. [6].

Third, the system gracefully degrades when components are unavailable. Without an OpenAI API key, the narrator falls back to a mock narration generator that reports YOLO detections verbally, and pyttsx3 provides fully offline TTS. This makes the system runnable for demonstration and testing without cloud dependencies.

C. Mapping to Beyond Vision

Each module in the laptop prototype corresponds to a specific hardware target in the Beyond Vision eyeglasses, summarized in Table I. The laptop webcam stands in for ArduCam IMX477 cameras, the CPU-bound YOLO inference for Hailo-8 accelerator inference, the cloud GPT-4V call for an eventual on-device VLM, and the laptop speaker for bone-conduction speakers.

TABLE I. Mapping from Software Prototype to Beyond Vision Hardware

Prototype Module	Beyond Vision Target
Laptop webcam	ArduCam IMX477 stereo
YOLOv8 on CPU	YOLO on Hailo-8 NPU
GPT-4V via cloud API	On-device compact VLM
OpenAI TTS / pyttsx3	Sherpa-ONNX offline TTS
Laptop speakers	Bone-conduction speakers

IV. IMPLEMENTATION

A. Software Stack

The system is implemented in a single Python 3.10 file (scene_narrator.py) totaling approximately 570 lines and depending on five external packages: ultralytics for YOLOv8 inference, opencv-python for camera capture and rendering, openai for GPT-4V and TTS API access, pyttsx3 for offline fallback TTS, and numpy as a transitive dependency. The codebase is organized into four classes (YOLODetector, SceneNarrator, TTSEngine, and SceneNarratorApp) plus a single Config class that exposes all tunable parameters in one location.

B. YOLO Detection Module

The YOLODetector class wraps the Ultralytics YOLOv8-nano model with a 0.40 confidence threshold. The nano variant was chosen for its favorable speed-accuracy tradeoff on CPU-only inference, where larger variants (s, m, l, x) introduce prohibitive latency. The detector outputs a list of dictionaries containing class label, confidence, and bounding box, and provides a format_detections method that groups detections by class for inclusion in the VLM prompt (e.g., “2 person(s) [conf: 0.85, 0.72]; 1 chair(s) [conf: 0.66]”).

C. GPT-4V Narration Module

The SceneNarrator class encodes each frame as a base64 JPEG (quality 85) and submits it to the gpt-4o endpoint with detail set to “low”, which lowers both latency and token cost while remaining sufficient for scene-level description. The system prompt instructs the model to act as a GPS-style narrator for a visually impaired user, to limit output to two or three short sentences, to avoid introductory phrases such as “I see” or “The image shows”, and to focus on spatial layout, notable objects, people, and hazards. The user message contains both the YOLO detection summary and the encoded image. Max output tokens are capped at 200 to enforce concision.

D. TTS Module

The TTSEngine class supports two backends. By default it uses the OpenAI tts-1 endpoint with the “nova” voice, streams the response to a temporary MP3 file, and plays it through platform-specific audio commands (afplay on macOS,

mpg123 or ffplay on Linux, default handler on Windows). If the OpenAI API is unavailable or the user disables it, the engine falls back to pyttsx3 with a rate of 160 words per minute, which uses the operating system’s native speech synthesizer and works fully offline.

E. Concurrency and User Interaction

Narration and TTS run on background daemon threads to avoid blocking the main capture and display loop, which must continue processing frames at the camera’s native rate. A simple `is_speaking` flag prevents overlapping speech, and a `narration_in_progress` flag prevents overlapping VLM calls. The user interface is rendered directly into the OpenCV display window and includes the live frame with YOLO overlays, a status bar showing detection count and time until the next narration, the most recent narration text, and keyboard controls (q to quit, s to force a narration, m to mute, l to print the log).

F. Logging

Each narration event is appended to an in-memory list containing the timestamp, detection count, YOLO summary string, and full narration text. On program exit the log is serialized to `narration_log.json`. This log file is the primary data source for the quantitative evaluation in Section VI.

V. EXPERIMENTAL SETUP

A. Hardware and Software Configuration

All experiments were conducted on a single commodity laptop running Ubuntu 22.04 with an Intel Core i7 CPU, 16 GB RAM, integrated graphics (no discrete GPU), and an internal 720p webcam. Python 3.10 was used with the package versions specified in `requirements.txt`. The OpenAI gpt-4o model was accessed via the cloud API over a residential broadband connection in Riyadh, Saudi Arabia. Background network conditions were not artificially controlled. The narration interval was set to five seconds, the YOLO confidence threshold to 0.40, and GPT-4V output was capped at 200 tokens with the image detail parameter set to “low”.

B. Recorded Sessions

We recorded two continuous sessions of the running system. Session 1 was a 130-second indoor recording in a single room, yielding 17 narration events. The camera observed a person at varying distances (including close-up frames with partial body occlusion), a smartphone held up to the camera displaying a cat photograph as the lock-screen wallpaper, a cabinet area with bottles and decorative items, and short periods of full and partial lens occlusion. Session 2 was a 297-second multi-room walkthrough during which the user moved through a hallway, kitchen, living room, and bedroom, encountering a real cat, cardboard boxes in transit zones, kitchen cabinetry, a TV and refrigerator, and a water dispenser. Session 2 produced 37 narration events. The two sessions together yield 54 events, which form the empirical basis for every claim in Section VI. Session 2 additionally records per-event end-to-end VLM latency, instrumented by wrapping the OpenAI API call with wall-clock timing.

C. Metrics

We analyze the combined session along five dimensions. (i) Detection composition: the distribution of YOLO detection counts across frames, used to characterize how often the detector contributes structured context. (ii) Detector failure modes: manual review of YOLO summaries against the corresponding narrations to identify false positives, miscounts, and miscategorizations. (iii) VLM behavior under detector failure: whether GPT-4V echoed, ignored, hedged on, or actively corrected each detector error. (iv) Navigational utility: keyword-based and manual annotation of whether each narration provided actionable directive guidance (“navigate around”, “be cautious of”, “obstacle ahead”) versus purely descriptive content. (v) End-to-end VLM latency, recorded for the 37 events in session 2.

D. Annotation Procedure

Each narration was annotated against the corresponding YOLO summary string and, where ground truth was known, against the actual scene. False positives were defined as YOLO labels that the authors could verify did not correspond to a real object of that class in the frame. Navigational utility was annotated by searching narrations for an

initial set of keywords (“navigate”, “caution”, “obstacle”, “avoid”, “harm”, “watch out”, “path”, “are of”, “way through”), then manually reviewing every match to classify it as directive (gives a movement instruction) or non-directive (e.g., the negation “no hazards visible”, which contains a keyword but provides no instruction).

VI. RESULTS AND ANALYSIS

A. Detection Composition

Across the 54 combined narration events, the YOLO detection count followed the distribution shown in Table II. The detector produced exactly one detection on 40.7 percent of frames, two or more detections on 16.7 percent, and zero detections on 42.6 percent. The high proportion of empty detection sets is the first significant finding of this work and persists even in object-rich indoor scenes containing curtains, walls, doors, ceiling fixtures, cabinetry, and decorative items. The effect was substantially more pronounced in the multi-room walkthrough (session 2: 59 percent of frames with zero detections) than in the single-room session (session 1: 24 percent), suggesting that the COCO-trained YOLOv8-nano model is well-tuned to person-centric stationary scenes but produces silent frames during transit through hallways and along walls where the dominant visible objects (doors, archways, architectural features, ceiling moldings) fall outside its training distribution.

TABLE II. YOLO Detection Count Distribution (N = 54 Events, Both Sessions)

Detections per frame	Session 1 (n=17)	Session 2 (n=37)	Combined
0	4 (24%)	19 (51%)	23 (42.6%)
1	9 (53%)	13 (35%)	22 (40.7%)
2+	4 (24%)	5 (14%)	9 (16.7%)

B. VLM Latency

Session 2 recorded per-event end-to-end VLM latency. Across the 37 events the latency was strongly right-skewed, with a median of 2.81 seconds and a mean of 3.20 seconds, but with a tail

extending to 8.89 seconds on a small number of events. The p95 of 8.70 seconds is dominated by what appear to be transient network conditions, since these events occurred non-contiguously and were not concentrated in any particular scene. Table III reports the distribution. The latencies fall within the five-second narration interval for the median event but exceed it in roughly 8 percent of events, occasionally causing a narration to arrive just as the next interval is starting.

TABLE III. VLM End-to-End Latency Distribution (N = 37, Session 2)

Statistic	Value (s)	Notes
Min	1.86	
Median	2.81	Within 5 s window
Mean	3.20	
p95	8.70	Exceeds 5 s window
Max	8.89	

C. Detector Failure Modes and VLM Correction

Manual review of both session logs surfaced three qualitatively distinct YOLO failure modes, summarized in Table IV. In every observed instance the VLM either ignored, rejected, or hedged on the detector’s label rather than echoing it.

Person double-counting. At three events in session 1 (17:14:43, 17:14:53, 17:15:57), YOLO reported two person detections when only one person was present. The second detection consistently carried lower confidence (0.59, 0.58, 0.42). The VLM’s narration in each case referred to a single person, implicitly resolving the duplicate by visually verifying the count.

Categorical misclassification. Five events across both sessions show this pattern. The session 1 cat-wallpaper event (17:15:09): YOLO reported “1 cat (0.88)” when the cat was a photograph on a phone screen; the VLM produced “phone with a wallpaper image of a cat.” At session 2 17:32:31, YOLO reported “1 cat (0.43)” and the VLM responded “no clear indications of a cat present in the shot.” At session 2 17:32:47, YOLO reported

“1 toilet (0.47)” in a hallway; the VLM responded “No toilets are visible, despite being detected with low confidence,” explicitly referencing and rejecting the detector’s call. At session 2 17:34:02, YOLO reported “1 person (0.64)” in an empty kitchen; the VLM’s narration silently described an empty kitchen with no person. At session 2 17:36:44, YOLO reported “1 bed (0.42)” and the VLM produced the notably hedged “There is a bed with a low confidence detection in the room,” appearing to fold the YOLO confidence value into its own epistemic language.

Silent frames. On 23 frames across both sessions YOLO returned an empty detection set. The VLM did not default to silence; instead it produced rich descriptions such as “You are indoors, facing a tall window with patterned curtains partially covering it. Light is coming through the window, illuminating a wooden cabinet nearby” (session 1, 17:16:27) and “You’re in a kitchen with brown cabinets. The upper cabinet is open, revealing dishes inside” (session 2, 17:33:38). In a strict ablation sense, more than 40 percent of the user-facing narrative content of the system originated from VLM perception that the detector did not contribute to.

TABLE IV. Detector Failure Modes and VLM Responses (Both Sessions)

Failure mode	Events	VLM response pattern
Person double-count	3	Narrated one person; ignored duplicate
Cat wallpaper → cat	2	Reframed as “phone wallpaper”
Spurious cat detection	1	“No clear indications of a cat”
Phantom toilet	1	“No toilets are visible, despite being detected”
Phantom person	1	Silently described empty scene
Low-conf bed	1	Hedged: “with a low confidence detection”

Zero detections	23	Produced rich, scene-grounded narration
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D. Confidence-Modulated Hedging

Two events in particular reveal that GPT-4V is not merely consuming the YOLO summary as a list of labels but reading and conditioning on the confidence values themselves. At session 2 17:32:47 the VLM’s narration explicitly contained the phrase “despite being detected with low confidence,” directly quoting the meaning of the 0.47 score in the prompt. At session 2 17:36:44 the VLM produced “There is a bed with a low confidence detection in the room,” again surfacing the 0.42 confidence value as part of its own epistemic language. This behavior was not requested in the system prompt and is an emergent property of supplying structured detector metadata to a capable VLM. It has practical implications: with prompt-level encouragement, the VLM could be made to communicate detector uncertainty to users systematically, rather than presenting every detection with equal apparent confidence.

E. The Navigation-Guidance Gap

The system’s stated design goal is to serve as a real-time navigation aid for visually impaired users. Of the 54 narration events, only 2 contained actionable directive guidance — sentences that told the user how to move through the environment. At session 2 17:34:40, in a hallway with a cardboard box, the VLM produced: “A potted plant is near a water dispenser in a room. There is a cardboard box in the foreground, and a window with curtains in the background. Make sure to navigate around the box if passing through.” At session 2 17:35:50 the system warned of glare: “Be cautious of the light as it may cause glare.” The remaining 52 events were descriptive rather than directive: they reported what was in the scene but did not tell the user what to do about it. Four additional events contained navigation-related keywords but were non-directive in substance, taking the form of negations (“no visible hazards,” “no obstacles or people are present,” “no immediate hazards are visible,” “The scene appears free of any immediate obstacles”); see Table V.

Quantitatively this is the most important finding of our evaluation: a system designed as a navigation aid produces navigation guidance on 3.7 percent of frames. Several mechanisms appear to contribute. The system prompt instructs GPT-4V to focus on “spatial layout, notable objects, people, and hazards,” but does not require directive output. The 200-token cap encourages concise scene reports rather than longer guidance. And the VLM’s training distribution likely skews toward descriptive image captioning rather than instructional pedestrian guidance. Crucially, in scenes where directive guidance was warranted (the hallway-with-box at 17:34:40 demonstrates the system is capable of producing it when prompted by the scene), the model still defaulted to description in the surrounding events. We discuss prompt-level and architectural mitigations in Section VII.

TABLE V. Navigation-Related Narrations Across Both Sessions

Timestamp	Class	Excerpt
17:34:40	Directive	“Make sure to navigate around the box if passing through.”
17:35:50	Directive	“Be cautious of the light as it may cause glare.”
17:15:39	Non-directive	“There are no visible hazards.”
17:33:16	Non-directive	“No obstacles or people are present.”
17:35:04	Non-directive	“No immediate hazards are visible.”
17:35:35	Non-directive	“The scene appears free of any immediate obstacles.”

F. Cold-Start and Disorientation Failures

The first event of session 2 (17:32:20) illustrates a distinct failure mode that did not appear in session 1. With the camera angled at a ceiling at session start, the VLM produced 50 words of description (“blurred image of what seems to be an

indoor ceiling and part of a wall... crown molding... door frame... light fixture”) but zero useful information for a visually impaired user. The narration is accurate as image captioning but is functionally useless: no person in motion benefits from being told that a ceiling is overhead. A related pattern appears on three blurry frames (17:34:09, 17:35:11, 17:36:52) where the VLM correctly diagnosed image quality issues but offered no guidance about what the user should do. These cold-start and disorientation cases argue for a meta-level layer that detects when narration is unlikely to be useful and either suppresses speech or asks the user to reorient the camera.

G. Behavior Under Occlusion

We tested two additional out-of-log occlusion conditions. With the camera fully covered, the VLM produced epistemically humble narrations: “The image is mostly dark with very little visible detail. There may be a faint light or mild texture in the lower part, but nothing distinct,” and “The image is mostly a dark brown gradient with no discernible objects or features visible.” With the camera partially covered by a finger, the VLM correctly localized the obstruction: “A person is close to the camera, looking downward. The image is slightly obstructed by a finger in the foreground on the left.” In neither case did the VLM fabricate objects to fill the visual void, a non-trivial result given the well-documented hallucination behavior of GPT-4V in low-information settings [5].

H. Representative Narrations

Table VI reproduces six representative narrations spanning the operating regimes observed in the combined sessions. They illustrate the five behavioral patterns documented in this section: competent person narration, detector disambiguation, silent-frame perceptual recovery, the rare directive guidance case, the cold-start failure, and graceful failure under full occlusion.

TABLE VI. Representative Narrations Across Operating Regimes

Regime	Verbatim narration
Standard person	“One person is sitting indoors facing forward. They are

	wearing glasses and a light shirt. The setting appears to be a room with light-colored walls and curtains in the background.”
Detector disambiguation	“No toilets are visible, despite being detected with low confidence. The area appears to be indoors with neutral-colored walls.”
Silent-frame recovery	“You’re in a kitchen space. There are dark wooden cabinets on the walls, some of which are open with visible items inside. A ceiling light is illuminating the area.”
Directive guidance	“A potted plant is near a water dispenser in a room. There is a cardboard box in the foreground... Make sure to navigate around the box if passing through.”
Cold-start failure	“You are looking at a blurred image of what seems to be an indoor ceiling... a section of crown molding and possibly a door frame... The image is quite out of focus.”
Full occlusion	“The image is mostly a dark brown gradient with no discernible objects or features visible. It appears to be very dimly lit or out of focus.”

VII. DISCUSSION, LIMITATIONS, AND FUTURE WORK

A. What the System Does Well

Across 54 events, the pipeline functioned end-to-end without intervention and produced narrations that a sighted reviewer judged consistently human-like, scene-grounded, and appropriately concise. Two findings are robustly supported by the data. First, the corrective behavior documented in Section VI-C is bidirectional: the VLM is not a passive consumer

of detector output but an active semantic checker that rejected every observed YOLO categorical error in the combined sessions. For an assistive deployment, where the cost of confidently announcing a non-existent toilet or cat to a blind user is non-trivial, this corrective behavior is arguably the most important practical benefit of the hybrid design. Second, the VLM’s emergent use of YOLO confidence values in its narration (Section VI-D) suggests that supplying structured detector metadata to a capable VLM transmits epistemic information that the model incorporates into its output, opening a route to principled uncertainty communication in future systems.

B. The Navigation-Guidance Gap

The most consequential weakness identified is the disparity between the system’s stated purpose and its observed output. A scene narrator intended to support visually impaired navigation produced directive guidance on 3.7 percent of frames (2/54), with the remaining 96.3 percent being scene description. The two directive narrations show the model is capable of guidance when the scene contains a clear navigational hazard (a cardboard box in a hallway, screen glare), but the default behavior is descriptive. For a deployed assistive product this rate is too low: a user walking through unfamiliar space needs to know what to do, not only what is around. The first event of session 2 illustrates a related failure: the system produced 50 words about a blurry ceiling and crown molding, none of which is actionable for a person in motion. We see three plausible mitigations. The first is prompt-level: explicitly require the model to end every narration with a directive sentence (“Suggested action:...”) or to suppress narration when no useful guidance is available. The second is architectural: layer a faster YOLO-driven obstacle-alert channel beneath the VLM narration so that immediate hazards trigger short imperative warnings (“Obstacle one meter ahead”) independently of the five-second VLM cadence. The third is data-side: fine-tuning a smaller VLM on assistive-narration data would likely shift the model’s default register from image captioning to pedestrian guidance. We expect that combining all three will be necessary; none is sufficient alone.

C. Other Limitations

The evaluation has several additional limitations. The 54 events came from two sessions in a single household with a single user; we cannot make population-level claims about failure-mode frequencies. The system depends on a cloud VLM, and the p95 latency of 8.7 seconds exceeds the five-second narration interval in a small but non-zero fraction of events, occasionally causing narrations to arrive late or overlap with the next interval. We evaluated only English narration, whereas the intended Beyond Vision deployment in Saudi Arabia is bilingual (Arabic and English) and the model’s right-to-left spatial language has not been tested. The evaluation was performed by sighted team members; no BLV participant has yet used the system, and the navigation gap identified here may be even more consequential to actual users than it appears in our review. Two implementation issues surfaced during session 2 that should be noted: the OpenAI TTS code path writes to a hardcoded `/tmp/scene_narration.mp3` location that does not exist on Windows, causing TTS playback (but not narration generation) to fail; and the streaming API call uses a deprecated method. Neither affects the textual log analysis presented here, but both should be fixed before any user deployment.

D. Future Work

Several extensions follow from this work. (1) Hardware port: the next step is moving the pipeline to the Beyond Vision hardware, replacing YOLO-on-CPU with YOLO-on-Hailo and evaluating whether the corrective and confidence-hedging VLM behaviors survive the transition to a smaller on-device VLM. (2) Directive-first prompt redesign: a targeted experiment varying the system prompt to require guidance-oriented output, with the navigation-keyword rate as the primary metric. (3) Dual-channel architecture: a fast detector-driven alert channel layered beneath periodic VLM narration to close the navigation-guidance gap. (4) Arabic-language evaluation: re-running both sessions with the system prompt rewritten in Arabic to test multilingual deployment quality. (5) Grounding ablation: rerunning the cat-wallpaper and toilet-false-positive scenes with the YOLO summary stripped from the prompt, to isolate how

much of the corrective behavior depends on the structured grounding rather than on the raw image alone. (6) BLV user study: longitudinal in-the-wild evaluation with blind and low-vision participants, with particular attention to whether the navigation gap translates into observed task failure in real environments.

VIII. CONCLUSION

We presented and empirically evaluated a hybrid YOLOv8 + GPT-4V + TTS pipeline for real-time audio scene narration intended for visually impaired users, with 54 narration events analyzed across two recorded sessions. The data supports two positive findings and exposes one critical gap. Positively, the value of detector-VLM hybridization in our sessions was bidirectional: in every observed instance of YOLO categorical error — phantom cat, phantom toilet, phantom person, low-confidence bed — GPT-4V rejected or hedged on the label rather than echoing it, and on the 42.6 percent of frames where the detector returned no detections the VLM produced rich, scene-grounded narration on its own perceptual capability. Two events further showed the VLM incorporating YOLO confidence values into its narrative language, an emergent route to user-facing uncertainty communication. Critically, however, the system gave actionable navigation guidance on only 2 of 54 events. For a system whose stated purpose is assistive navigation, this gap between scene description and movement instruction is the central practical obstacle to deployment, and closing it is the most important task remaining. The implementation closes the practical gap identified in our prior literature review and serves as a software proof-of-concept for the Environmental Awareness module of the Beyond Vision assistive eyeglasses platform; the hardware port, a directive-first prompt redesign, a dual-channel detector-alert architecture, Arabic evaluation, and a BLV user study constitute the next phases of the project.

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